Problem set 2

Solution Key

Econ 5030

Due 10/12/26

Payoff pairs are written (row player, column player) unless stated otherwise.

1 Iterated deletion

	n_P	l_P	h_P
n_G	$0, 0$	$1, 2$	$1, 2$
l_G	$3, 1$	$3, 3$	$3, 4$
h_G	$2, 1$	$4, 3$	$5, 5$

1.A) *Round 1.* For Greg, n_G is strictly dominated by l_G ($3 > 0,\ 3 > 1,\ 3 > 1$). For Phil, n_P is strictly dominated by l_P ($2 > 0,\ 3 > 1,\ 3 > 1$). (h_P only *weakly* dominates l_P, since $2 = 2$ against n_G, so l_P is not deleted yet.) Delete n_G and n_P.

Round 2. In the remaining game

	l_P	h_P
l_G	$3, 3$	$3, 4$
h_G	$4, 3$	$5, 5$

h_G strictly dominates l_G ($4 > 3,\ 5 > 3$) and h_P strictly dominates l_P ($4 > 3,\ 5 > 3$). Delete l_G and l_P.

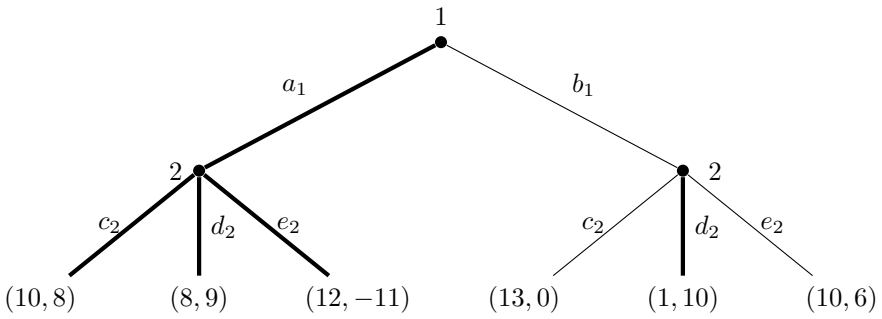
Nothing else can be deleted. The solution set is $\{h_G\} \times \{h_P\}$, i.e. the profile (h_G, h_P) with payoffs $(5, 5)$.

1.B) Best responses: $BR_G(n_P) = \{l_G\}$, $BR_G(l_P) = \{h_G\}$, $BR_G(h_P) = \{h_G\}$; $BR_P(n_G) = \{l_P, h_P\}$, $BR_P(l_G) = \{h_P\}$, $BR_P(h_G) = \{h_P\}$.

The only profile where each strategy is a best response to the other is (h_G, h_P). So the set of Nash equilibria is $\{(h_G, h_P)\}$. Since the game is solved by iterated strict dominance, this is also the unique equilibrium in mixed strategies: a strategy that is deleted by iterated strict dominance gets zero probability in every Nash equilibrium.

2 Putting games into the Normal Form

1. Game tree. Payoffs are (u_1, u_2). Thick branches are the moves chosen by backward induction (part 5).



2. Strategies. $S_1 = \{a_1, b_1\}$. Player 2 moves at two information sets (after a_1 and after b_1), so a strategy is a pair (x, y): play x after a_1 and y after b_1, with $x, y \in \{c_2, d_2, e_2\}$:

$$S_2 = \{c_2c_2,\ c_2d_2,\ c_2e_2,\ d_2c_2,\ d_2d_2,\ d_2e_2,\ e_2c_2,\ e_2d_2,\ e_2e_2\} \quad (3 \times 3 = 9 \text{ strategies}).$$

3. Normal form (rows Player 1, columns Player 2; best responses underlined).

	c_2c_2	c_2d_2	c_2e_2	d_2c_2	d_2d_2	d_2e_2	e_2c_2	e_2d_2	e_2e_2
a_1	$10, 8$	$\underline{10}, 8$	$\underline{10}, 8$	$8, \underline{9}$	$\underline{8}, \underline{9}$	$8, \underline{9}$	$12, -11$	$\underline{12}, -11$	$\underline{12}, -11$
b_1	$\underline{13}, 0$	$1, \underline{10}$	$\underline{10}, 6$	$\underline{13}, 0$	$1, \underline{10}$	$\underline{10}, 6$	$\underline{13}, 0$	$1, \underline{10}$	$10, 6$

4. Nash equilibria. Against a_1, Player 2's best responses are the strategies with d_2 after a_1 ($9 > 8 > -11$): d_2c_2, d_2d_2, d_2e_2. Against these, a_1 gives Player 1 8, while b_1 gives $13, 1, 10$; only d_2d_2 keeps a_1 optimal. Against b_1, Player 2's best responses have d_2 after b_1: c_2d_2, d_2d_2, e_2d_2. Against these, b_1 gives 1 but a_1 gives $10, 8, 12$, so there is no equilibrium with b_1.

The unique pure-strategy Nash equilibrium is $(a_1,\ d_2d_2)$, with outcome (a_1, d_2) and payoffs $(8, 9)$.

(Mixed strategies: Player 1 cannot mix, because if both a_1 and b_1 have positive probability Player 2's unique best response is d_2d_2, and then a_1 is strictly better. With Player 1 playing a_1, Player 2 can play d_2 after a_1 and randomize after b_1 with probabilities (q_c, q_d, q_e), provided $13q_c + q_d + 10q_e \leq 8$. Every such equilibrium has the same outcome (a_1, d_2).)

5. Incredible threats? No. By backward induction Player 2 plays d_2 after a_1 ($9 > 8 > -11$) and d_2 after b_1 ($10 > 6 > 0$). Player 1 then compares 8 (from a_1) with 1 (from b_1) and plays a_1. The subgame perfect equilibrium is (a_1, d_2d_2), which is exactly the unique pure Nash equilibrium. What deters b_1 is Player 2's response d_2, and d_2 is her best move once b_1 has been played, so the threat is credible. (In the mixed equilibria above, Player 2's off-path weight on c_2 or e_2 is not optimal. But those moves would give Player 1 13 or 10, which makes b_1 *more* attractive, so they are not threats that hold the equilibrium together.)

3 Timing and Strategies

2.1 Simultaneous. Neither player observes the other's choice, so each player has a single information set and a strategy is just an action: $S_1 = S_2 = \{0, 1\}$.

P1\P2	0	1
0	$1, 0$	$0, 1$
1	$0, 1$	$1, 0$

This is Matching Pennies. There is no pure Nash equilibrium. The unique equilibrium has each player choosing 0 and 1 with probability $\frac{1}{2}$, so each expects $\$\frac{1}{2}$.

2.2 Sequential. Player 1 still has $S_1 = \{0, 1\}$. Player 2 moves after seeing Player 1's choice, so she has two information sets and her strategy must be a complete contingent plan (x, y): play x if Player 1 chose 0 and y if Player 1 chose 1:

$$S_2 = \{(0,0),\ (0,1),\ (1,0),\ (1,1)\} = \{\text{always 0},\ \text{copy},\ \text{do the opposite},\ \text{always 1}\}.$$

2.3 Difference in the normal forms. The physical actions are the same, but the strategy sets are not. The simultaneous game is 2×2. The sequential game is 2×4:

P1\P2	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
0	$1,0$	$1,0$	$0,1$	$0,1$
1	$0,1$	$1,0$	$0,1$	$1,0$

Timing and information enter the normal form only through the strategy sets. Because Player 2 can condition on what she sees, she has more strategies, and some payoff columns repeat. This changes the predictions. In the sequential game the strategy $(1, 0)$ (do the opposite) wins for Player 2 against anything. The Nash equilibria are Player 2 playing $(1, 0)$ with Player 1 playing anything; in pure strategies these are $(0, (1, 0))$ and $(1, (1, 0))$, with payoffs $(0, 1)$. In the simultaneous game each player wins with probability $\frac{1}{2}$. The 2×4 matrix would also describe a simultaneous game in which Player 2 submits a contingent plan in advance, so the normal form records information only through what strategies can condition on.

4 Third party observation of a game

2.4 No, not by itself. If each player's payoff were her own money, (M_1, M_2) would not be a Nash equilibrium: holding the opponent at M, a player gains by deviating. The table does not label whose dollar amount comes first, but the conclusion holds under either reading:

- First entry = row player (Player 2). Against M, switching to L pays \$30 > \$18. The money game's equilibria are (L_1, M_2), (M_1, L_2), and a mixed equilibrium with each player choosing L with probability 6/7 and M with probability 1/7.
- First entry = Player 1. Against M, switching to H pays \$20 > \$18. Here L is strictly dominated by M, and once L is deleted, M is strictly dominated by H, so the unique equilibrium is (H_1, H_2) with \$5 each.

So the data reject the joint hypothesis that players maximize their *own money* and play a Nash equilibrium. They do not reject game theory. In a game, payoffs are utilities that represent each player's preferences over outcomes. The experimenter controls the money but not the preferences, and if players care about more than their own earnings, the game they are actually playing has different payoffs from the money table. Universal M play is consistent with Nash play of such a game (2.6).

2.5 Normal form. A normal-form game is $G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$, where:

- N is the set of players;

- S_i is Player i's set of strategies, where a strategy is a complete contingent plan specifying an action at every point where i might move, including points that are never reached;
- $u_i : S \to \mathbb{R}$, with $S = \prod_j S_j$, gives i's payoff from each strategy profile. It is a utility representation of i's preferences over the outcome the profile produces. It is ordinal for pure-strategy analysis, and once mixed strategies or beliefs are involved it must be a von Neumann–Morgenstern utility.

The normal form treats players as choosing whole strategies simultaneously and independently. Timing and information are not shown directly; they show up only through what the strategies can condition on. In this experiment $N = \{1, 2\}$ and $S_i = \{L_i, M_i, H_i\}$, but u_i is the utility a player gets from the money pair, which need not equal her own money.

2.6 Explanation and game. Postulate social (fairness) preferences: each subject values the pair's total earnings and puts extra weight on the worse-off subject's earnings (Charness–Rabin-style social-welfare preferences):

$$u_i(x_1, x_2) = x_1 + x_2 + \min\{x_1, x_2\}.$$

Because u_i is symmetric in (x_1, x_2), both players get the same utility in each cell, whichever way the money table is read. The normal form is

	L_1	M_1	H_1
L_2	$24, 24$	$50, 50$	$17, 17$
M_2	$50, 50$	$\mathbf{54, 54}$	$28, 28$
H_2	$17, 17$	$28, 28$	$15, 15$

For each player M strictly dominates both L and H: against L, $50 > 24$ and $50 > 17$; against M, $54 > 50$ and $54 > 28$; against H, $28 > 17$ and $28 > 15$. Iterated deletion leaves only (M_1, M_2), so it is the unique Nash equilibrium, in pure or mixed strategies. This matches the data.

Can these data test rationality? Not on their own. Rationality means choosing a best response under *some* complete, transitive preferences (and beliefs). The experimenter observes money and choices but not utilities. As 2.6 shows, any observed profile can be made the unique (even dominant-strategy) equilibrium by choosing preferences suitably, so choices in a single game cannot refute rationality. They only test rationality *jointly* with an assumed utility function (for example, "own money only") and equilibrium beliefs. A real test needs restrictions that tie preferences together across situations. For example, vary the payoffs across many games and check whether one stable utility function explains all the choices (a revealed-preference consistency test), or elicit beliefs and preferences separately.

5 Subgame Perfect Nash Equilibrium

(I read the timing as: Player 2 observes Player 1's action and then chooses from $\{U_2, D_2\}$, as the subscripts indicate.)

1. Normal form. $N = \{1, 2\}$, $S_1 = \{L_1, M_1, H_1\}$. Player 2 has three information sets, so $S_2 = \{U_2, D_2\}^3$, with $2^3 = 8$ strategies (x_L, x_M, x_H), where x_k is her action after k_1. Payoffs are $u_i(s_1, s_2) = u_i(s_1, x_{s_1})$.

2. Matrix and Nash equilibria (strategies written $x_L x_M x_H$; best responses underlined).

	UUU	UUD	UDU	UDD	DUU	DUD	DDU	DDD
L_1	$\underline{10}, 10$	$\underline{10}, 10$	$\underline{10}, 10$	$\underline{10}, 10$	$1, \underline{11}$	$1, \underline{11}$	$1, \underline{11}$	$1, \underline{11}$
M_1	$5, 5$	$5, 5$	$2, \underline{6}$	$2, \underline{6}$	$\underline{5}, 5$	$\underline{5}, 5$	$\underline{2}, \underline{6}$	$2, \underline{6}$
H_1	$1, 4$	$3, \underline{5}$	$1, 4$	$3, \underline{5}$	$1, 4$	$3, \underline{5}$	$1, 4$	$\underline{3}, \underline{5}$

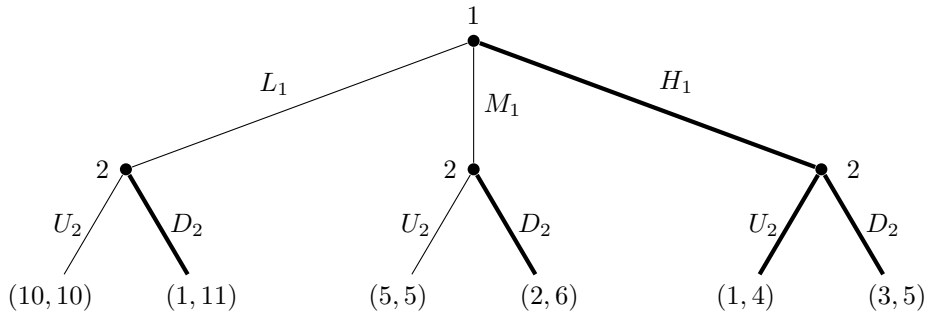
Player 2 strictly prefers D_2 after every action ($11 > 10$, $6 > 5$, $5 > 4$), so in equilibrium she plays D_2 at the node that is reached.

- L_1: Player 2 must play D after L, so Player 1 gets 1, but M_1 gives at least 2. No equilibrium.
- M_1: needs $x_M = D$; $x_L = D$ (else L_1 gives $10 > 2$); $x_H = U$ (else H_1 gives $3 > 2$). This gives (M_1, DDU).
- H_1: needs $x_H = D$; $x_L = D$ (else $10 > 3$); $x_M = D$ (else $5 > 3$). This gives (H_1, DDD).

The pure-strategy Nash equilibria are $(M_1,\ D_2D_2U_2)$ with payoffs $(2, 6)$ and $(H_1,\ D_2D_2D_2)$ with payoffs $(3, 5)$.

(Mixed strategies: Player 1 never mixes in equilibrium. There are also mixed versions of each equilibrium that differ only off the path. With the outcome (M_1, D_2): $\Pr(U \mid L) \leq 1/9$ and $\Pr(U \mid H) \geq 1/2$. With the outcome (H_1, D_2): $\Pr(U \mid L) \leq 2/9$ and $\Pr(U \mid M) \leq 1/3$.)

3. Game tree and SPE. Payoffs are (u_1, u_2). Thick branches are the backward-induction choices.



Backward induction: in each of the three subgames Player 2 plays D_2 ($11 > 10$, $6 > 5$, $5 > 4$). Anticipating this, Player 1 gets 1, 2, 3 from L_1, M_1, H_1 and chooses H_1.

The unique subgame perfect Nash equilibrium is $(H_1,\ D_2D_2D_2)$, with outcome (H_1, D_2) and payoffs $(3, 5)$. The other Nash equilibrium $(M_1, D_2D_2U_2)$ is not subgame perfect. It rests on the incredible threat to play U_2 after H_1, which Player 2 would not carry out because it gives her $4 < 5$.